

Train a CBOW model using the following sentence:

“The dog barks at night.”

The **window size is set to 2**.

The vocabulary consists of the following unique words:

[{the, dog, barks, at, night}]]

For simplicity, assume the CBOW model has already assigned the following **one-hot encoded vectors**:

Word One-hot Vector

the [1, 0, 0, 0, 0]

dog [0, 1, 0, 0, 0]

barks [0, 0, 1, 0, 0]

at [0, 0, 0, 1, 0]

night [0, 0, 0, 0, 1]

Tasks

(i) Identify the **context words** surrounding the center word **“barks”**.

(Left and right context based on window size)

(ii) Write the **context vectors** using the one-hot encodings.

(iii) Compute the **average context vector** by summing the context vectors and dividing by the number of context words.

Assume the following parameters:

Weights (Input → Hidden Layer)

Word	H1	H2	H3	H4
the	0.2	0.1	0.3	0.4
dog	0.4	0.3	0.1	0.2
barks	0.5	0.4	0.2	0.1
at	0.1	0.2	0.4	0.3
night	0.3	0.5	0.2	0.4

Bias for hidden layer = [0.1, 0.2, 0.3, 0.1]

Weights (Hidden → Output Layer)

Hidden	the	dog	barks	at	night
H1	0.3	0.2	0.4	0.1	0.5
H2	0.1	0.4	0.3	0.2	0.1
H3	0.2	0.3	0.1	0.5	0.4
H4	0.4	0.1	0.2	0.3	0.2

Bias for output layer = [0.1, 0.2, 0.1, 0.2, 0.3]

(iv) Compute the **hidden layer activations**.

(v) Compute the **output scores**.

(vi) Identify the **predicted center word**.

Train a Skip-Gram model using the following sentence:

“Students study machine learning.”

The **window size is set to 1**.

The vocabulary consists of:

[
 { {students, study, machine, learning} }
]

Assume the following **one-hot vectors**:

Word One-hot Vector

students [1, 0, 0, 0]

study [0, 1, 0, 0]

machine [0, 0, 1, 0]

learning [0, 0, 0, 1]

Tasks

(i) Identify the **center word** and its **context words** when the center word is “**study**”.

(ii) Write the **input vector** for the center word.

(iii) Compute the **hidden layer representation** using the given weights.

Assume the following parameters:

Weights (Input → Hidden Layer)

Word H1 H2 H3

students 0.2 0.3 0.1

study 0.4 0.2 0.5

machine 0.3 0.4 0.2

learning 0.1 0.5 0.3

Bias for hidden layer = **[0.1, 0.2, 0.1]**

Weights (Hidden → Output Layer)

Hidden students study machine learning

H1 0.3 0.2 0.4 0.1

H2 0.1 0.5 0.2 0.3

H3 0.4 0.1 0.3 0.2

Bias for output layer = **[0.1, 0.2, 0.1, 0.3]**

(iv) Compute the **output scores** for each context word.

(v) Identify the **most likely context word** predicted by the Skip-Gram model.